PHENIKAA UNIVERSITY

Faculty of Computer Science

CSE703016: Introduction to HCI (Giao Diện Người Máy)

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Today’s Topics

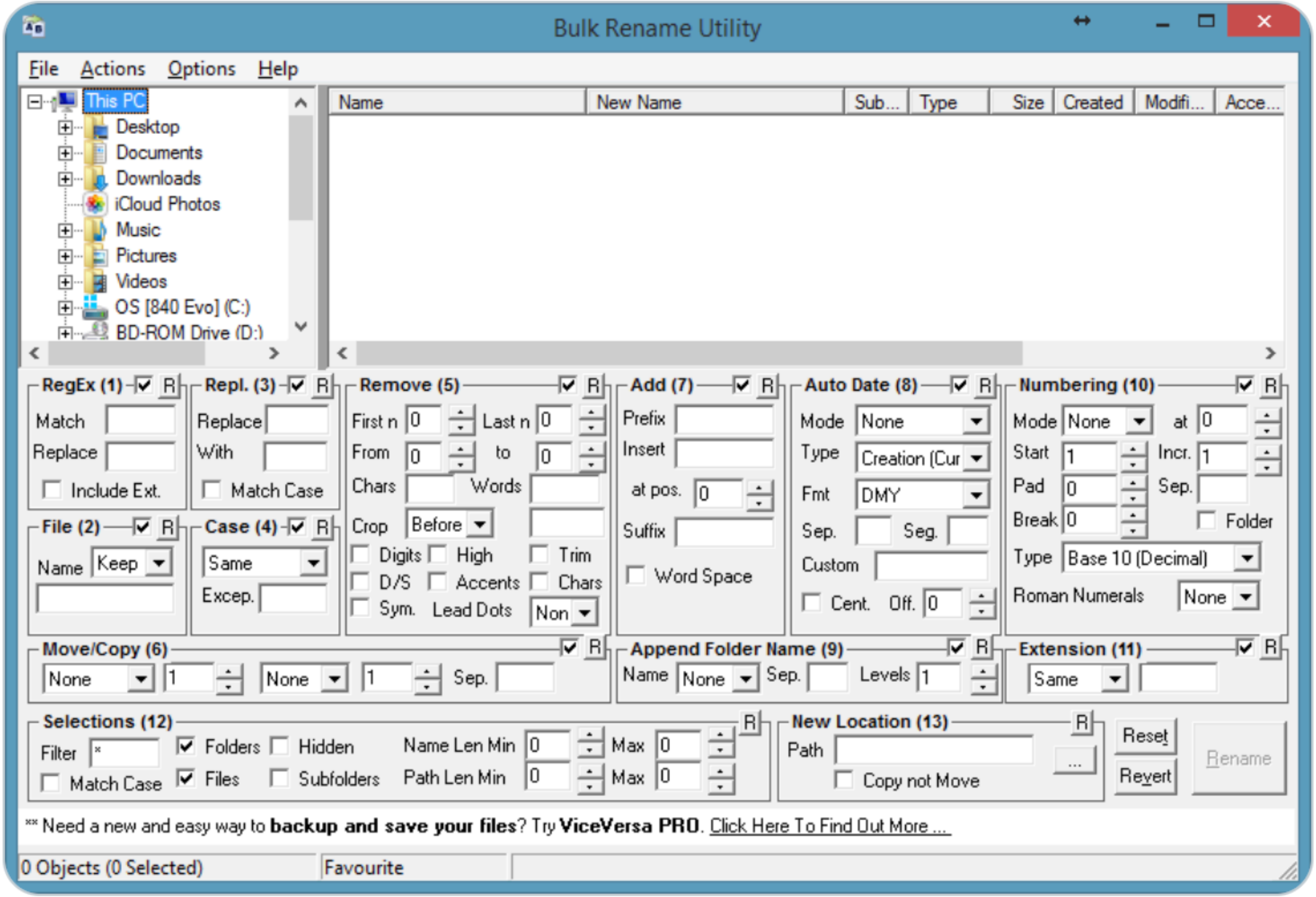
- UI Hall of Fame and Shame
- Introductions!
- What is HCI?
- Why is interaction design hard?
- Activity
- Course Overview

UI Hall of Shame

What are examples of truly terrible UIs you've seen in the wild?
[type in the chat]

John Bellomy
@cowbs

Engineers don't let engineers design user interfaces.



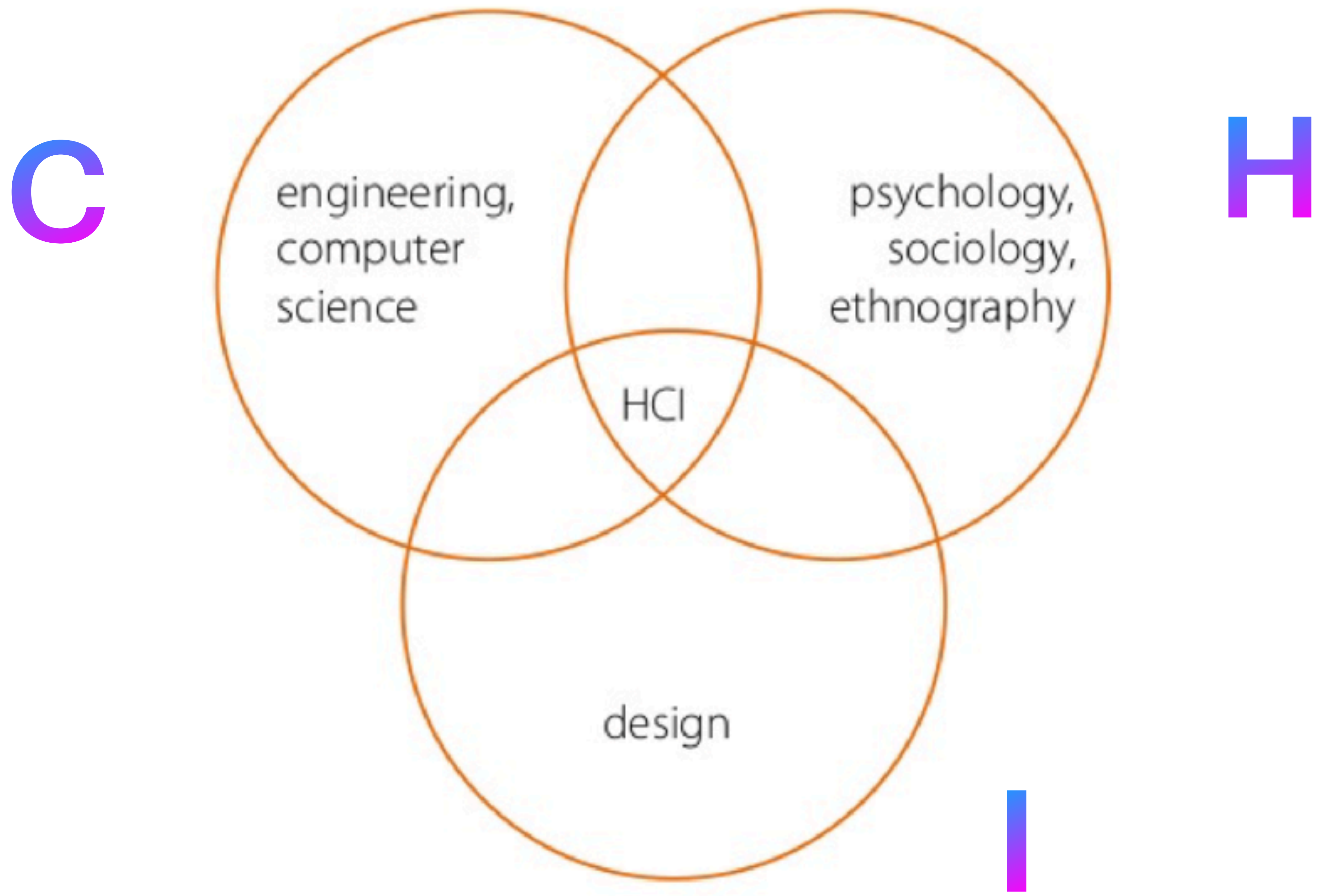
What’s wrong with this UI?
[type in the chat]

UI Hall of Fame

Nothing for today, but think about UIs that you really like for future classes!

What is HCI?

Human-Computer Interaction



HCI is about the **design and use** of computer technology

https://www.slideshare.net/agaszostek/history-and-future-of-human-computer-interaction-hci-and-interaction-design

What is **design**?

- Design is about **making** things
- "[Design is] a plan for arranging elements in such a way as to best accomplish a particular **purpose**." - Charles Eames *(designer of that Eames chair by Herman Miller)*

What is an **interface**?

- “the place at which independent and often unrelated systems meet and act on or communicate with each other” - Merriam Webster dictionary
- This is the **interaction** part of human-computer interaction
- The interface is what we design!

Why is interaction design hard?

You are not the user!

- User interfaces are about communicating with users. Users are NOT like you!
- As the builder, you know a lot more about your application than any user will, and it's difficult to un-learn it.
- What do we mean when we say "the user is always right"?

BUT don’t expect users to be designers either

1) Telephone handset weight

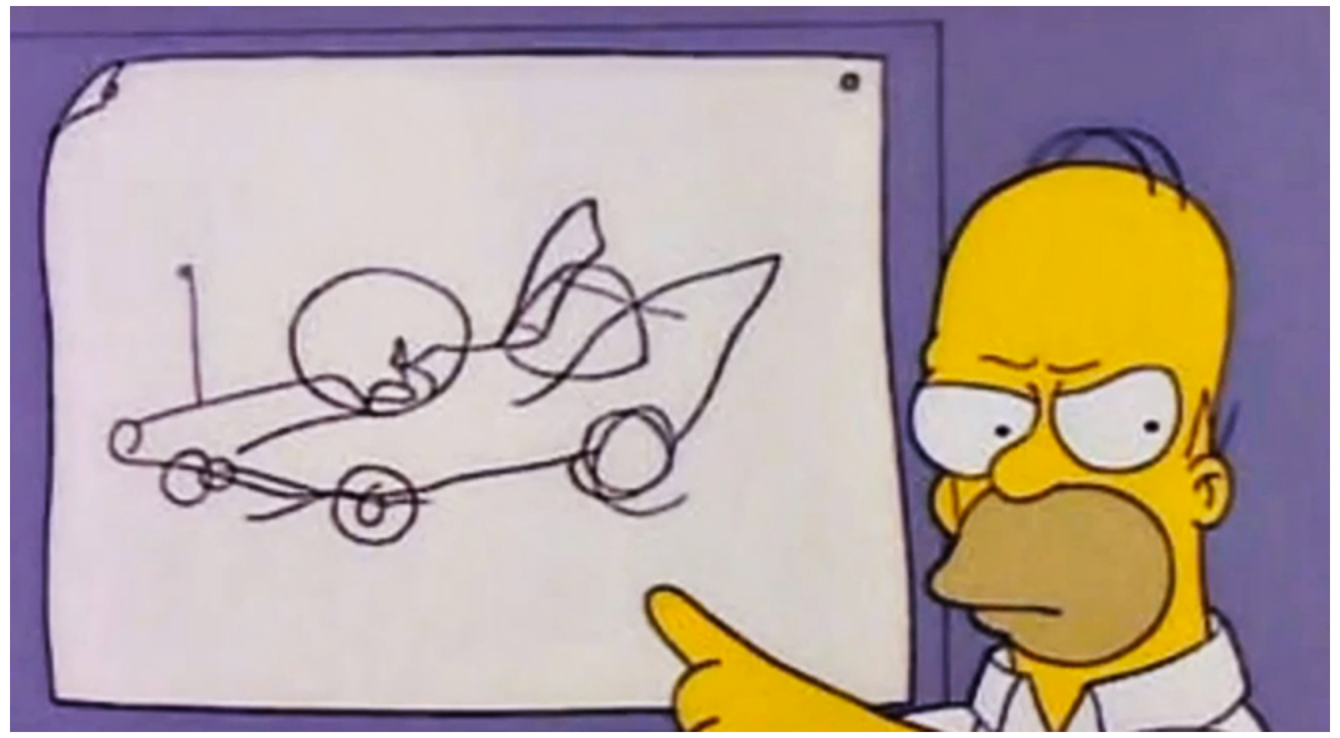
- users said: it’s fine! — but they wanted lighter

2) # of Google search results

- users said: 30 results — but they really wanted 10

3) Command abbreviations

- Users made 2x more errors with their own custom abbreviations



“I want a horn here, here, here, and here. You can never find a horn when you’re angry.”

1) Klemmer, Ergonomics, Ablex, 1989, pp 197-201

2) http://perspectives.mvdirona.com/2009/10/31/TheCostOfLatency.aspx

3) Grudin & Barnard, “When does an abbreviation become a word?”, CHI ’85

So how do you know what to design?

- Design as a **process**:
  - To synthesize a solution from all the relevant constraints
  - To frame, or reframe, the problem and objective
  - To create and envision alternatives
  - To select from those alternatives
  - To visualize and prototype the intended solution

- Bill Moggridge (co-founder of IDEO)

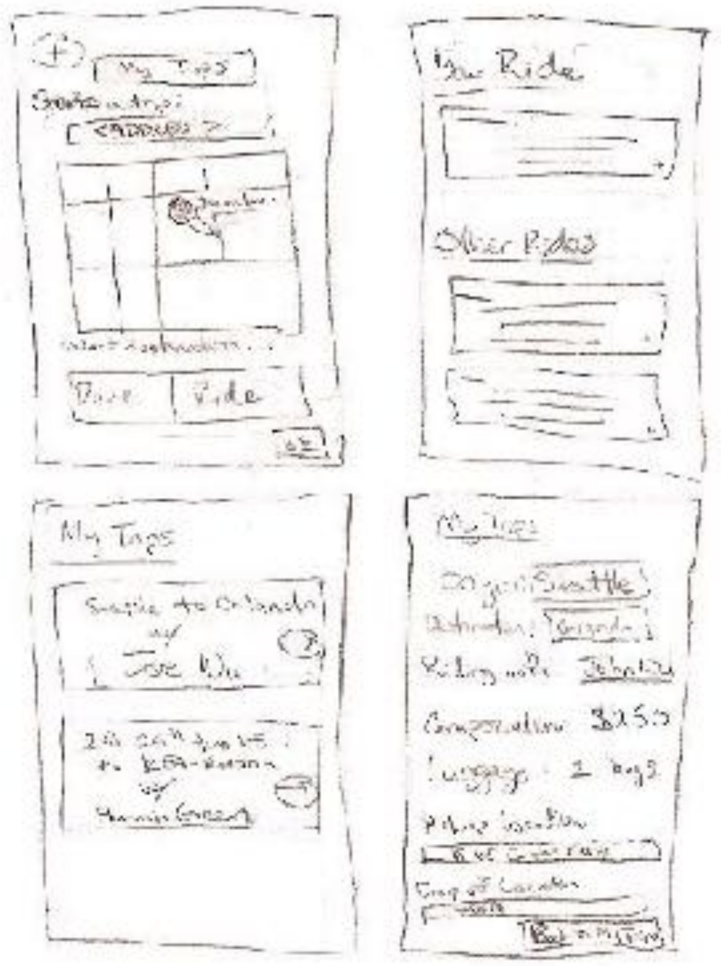
Design as a **process**

- Design process as **iterative** and **explorative**, constantly involving **users** and investigating **use**, since we can't just trust our instincts
- Group project - propose and carry out an end-to-end design process

Talk to people, investigate problems

→

Sketching and Storyboarding



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Low-fidelity Prototyping

→

Digital Mockup

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Presentation & Communication

So how do you know what to design?

- Design as an open-ended **series of principles**:
  - Usability: how well users can use the system's functionality
    - Learnability: how easy is it to learn?
    - Efficiency: once learned, how quickly can it be used?
    - Safety: are errors few and recoverable?
  - Accessibility
  - Aesthetics, minimalism
  - Ergonomics
  - Expressivity, flexibility
  - Malleability, control

Which principles to emphasize depends on context

- There are **trade-offs** between different design principles, so you can't just apply them mindlessly
- Emphasis depends on the **user**
  - Novice users need greater learnability
  - Expert users need efficiency
  - But everyone can be a novice or an expert at different points in time
- Emphasis depends on the **task**
  - Highly critical tasks should emphasize safety (amber alert system)
  - Less critical, repetitive tasks need efficiency (unlocking your smartphone)

There are also other trade-offs

- Software builders have a lot to worry about!
  - functionality
  - security
  - performance
  - maintainability
  - cost
  - reliability

- Some of your other CSE courses focus on these other attributes
- But we'll mostly ignore these trade-offs in this class in favor of how well the interface addresses a **problem that people have** and how successful is the interface for people to **achieve their goals and tasks**
- Just like with these other attributes, we have to think about constraints —but now, we add humans' physical, mental, and social constraints in addition to physical constraints of machines

Objectives of this Course

- **Process**-focused perspective on interaction design
- Learn **design principles** and **heuristics**
- Learn **design methods** such as **user** research methods, **task** analysis
- Create **design artifacts** for communicating and thinking through your intermediate design: scenarios, sketches, storyboards, prototypes
- Rapid prototyping and **iteration** of design, coupled with user testing and other evaluation methods
- **Critical** perspective on design solutions
- Learning communication skills and design **critique**

Activity

- Here are three different kinds of locks.
- Consider the different design decisions made in these different locks to address different principles (learnability, efficiency, safety)
- Since we don't have them in front of us, think back to a time where you used a lock, and remember problems you may have encountered when trying to use it.
- What aspect of the design of the lock related to the problem?
- Design and then sketch a new lock that addresses the problem.
- Upload your sketch to Canvas

Course Overview

Course Structure

- We have lectures and in-class activities, along with group work or group presentations with feedback sometimes.
- Practice section: Participate in group presentation and give feedback
- As you can see - **active participation** is a huge component of the course!
- Check Canvas for syllabus, assignments
- Turn in group assignments in Canvas

Useful Resources

- All about UI/UX design learning guide: https://github.com/hendurhance/ui-ux
- Good coursera course: https://www.coursera.org/learn/ui-design

Forming Groups

- Now take time to decide your teams
- Form teams on Canvas group
- **The teams we create today are final, no exception!**

Upcoming schedule

- Next lecture topic: The **Design Process**
- The teams we create today are final, no exception!
- You’ll also work with your team and start brainstorming for Assignment1a (on Canvas)

Q & A